

Engine

 [Ford Ecat Electronic Parts Catalog](#)

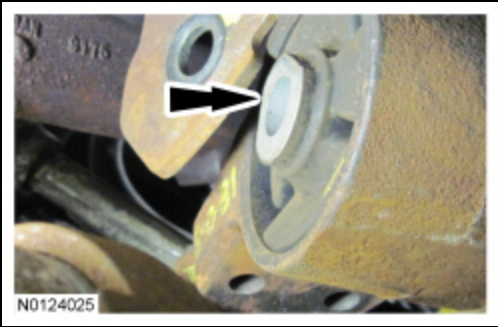
Special Tool(s)

 ST1377-A	Lifting Bracket, Engine 303-F047 (014-00073) or equivalent
 ST3110-A	Support Brackets, Engine 303-1507 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Material

Item	Specification
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS-M2C945-A
Motorcraft® Threadlock 262 TA-26	WSK-M2G351-A6

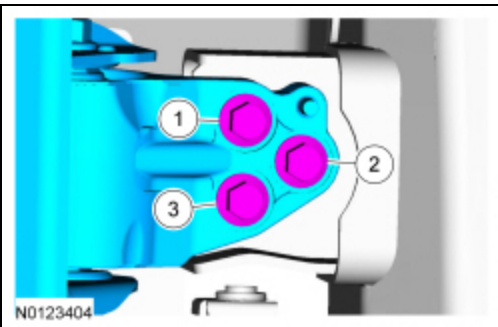
1. Using the Heavy Duty Floor Crane and the Engine Lifting Bracket, position the engine in the engine compartment leaving access to the LH engine support insulator bolts.
2. **NOTE:** *Clean the LH and RH engine support insulator-to-frame mating surfaces of any dirt or foreign material prior to installation.*
- Install the LH engine support insulator from under the vehicle.



3. **NOTE:** Apply threadlock to the bolt threads prior to installation.

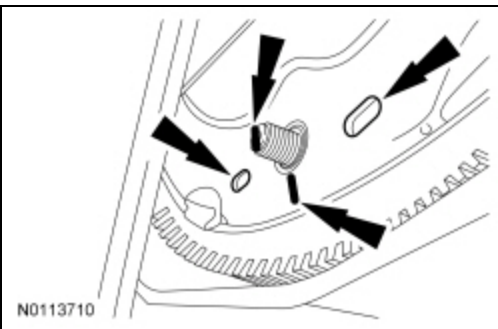
Using a suitable floor crane, raise the engine assembly to gain access to install the 3 LH engine support insulator bolts.

- Tighten the 3 LH engine support insulator bolts in the sequence shown to 175 Nm (129 lb-ft).



4. **NOTICE:** The torque converter must be installed in it's original orientation to the flexplate or engine damage may occur.

Check that the flexplate is positioned so that the 2 sloted holes are shown at the 6 o'clock position and the index marks are aligned.

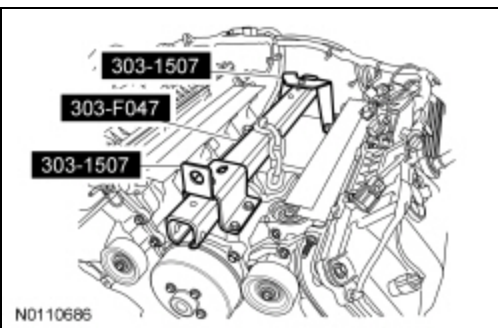


5. Install the LH exhaust manifold heat shield and bolts.

- Tighten to 12 Nm (106 lb-in).

6. Lower the engine.

7. Remove the Engine Lifting Bracket and Engine Support Brackets.



8. Connect the steering column shaft to the steering gear and install the bolt.

- Tighten to 30 Nm (22 lb-ft).

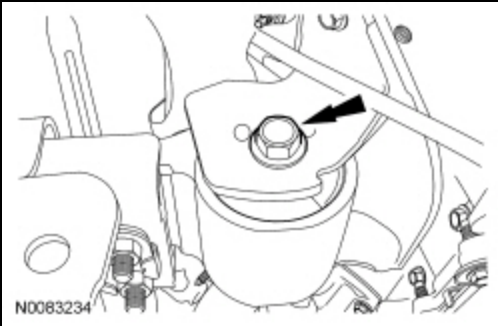
9. Install the coolant expansion tank and bolts.

- Tighten to 15 Nm (133 lb-in).

10. **NOTICE: Only use hand tools when tightening the engine support insulator through bolt or the engine support insulator may be damaged.**

Apply threadlock to the bolt threads and install the LH engine support insulator bolt.

- Tighten to 350 Nm (258 lb-ft).



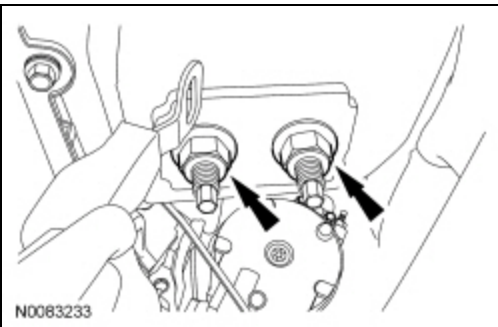
11. If equipped, install the front driveshaft. For additional information, refer to Section 205-01.

12. **NOTICE: Only use hand tools when tightening the engine support insulator nuts or the engine support insulator may be damaged.**

NOTE: Make sure the RH engine support insulator mating surfaces and the washer mating surface are free of foreign material and corrosion before installation. Install a new washer.

Apply threadlock to the stud threads and install the RH engine support insulator studs, washer and nuts.

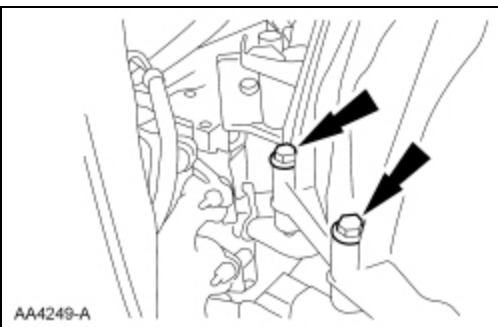
- Tighten studs to 15 Nm (133 lb-in).
- Tighten nuts to 175 Nm (129 lb-ft).



13. **NOTE:** The upper 2 transmission-to-engine bolts will be installed later.

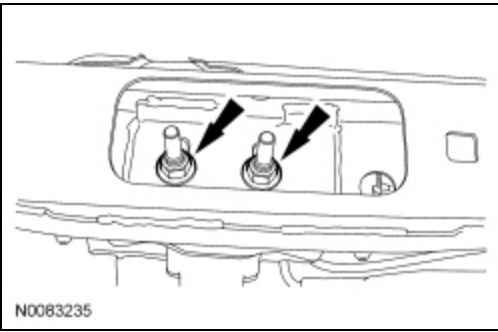
Install the lower 5 transmission-to-engine bolts.

- Tighten to 48 Nm (35 lb-ft).



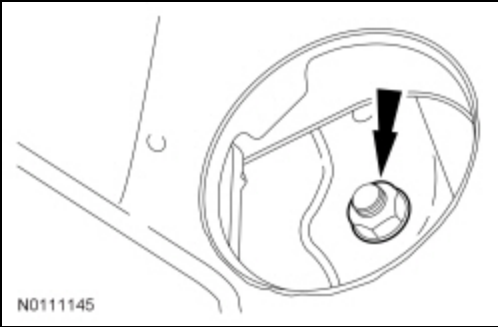
14. Tighten the 2 transmission insulator and retainer nuts.

- Tighten to 90 Nm (66 lb-ft).

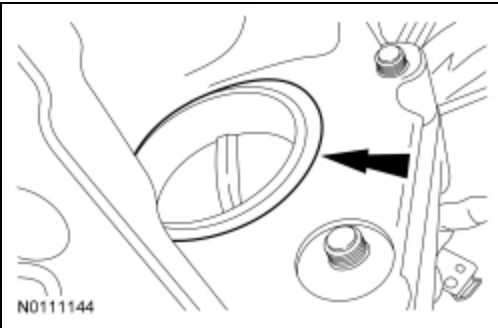


15. Install 4 new torque converter-to-flexplate nuts.

- Tighten to 40 Nm (30 lb-ft).

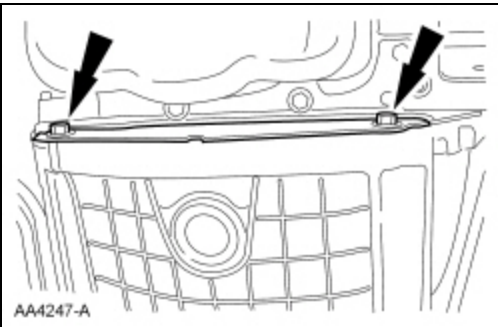


16. Install the cylinder block opening cover.



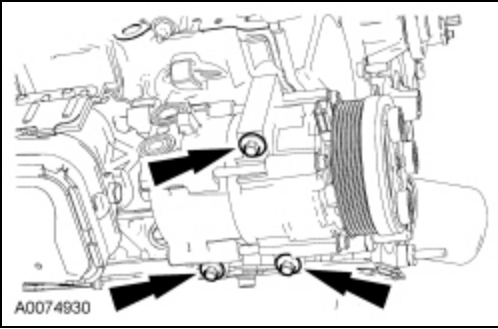
17. Install the flexplate inspection cover and the 2 bolts.

- Tighten to 48 Nm (35 lb-ft).

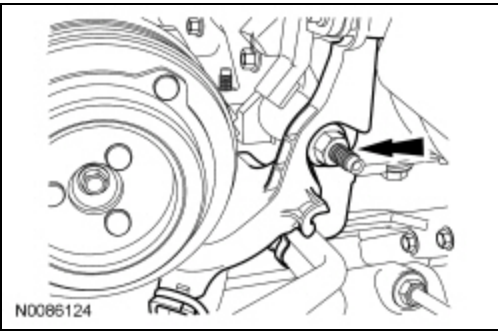


18. Position the A/C compressor and install the 3 bolts.

- Tighten to 25 Nm (18 lb-ft).

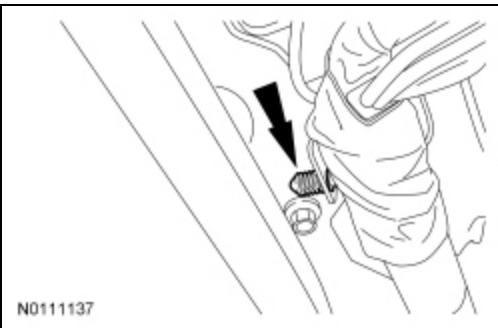


19. Position the transmission cooler tube support bracket, the starter wiring harness support bracket and install the nut.
- Tighten to 10 Nm (89 lb-in).

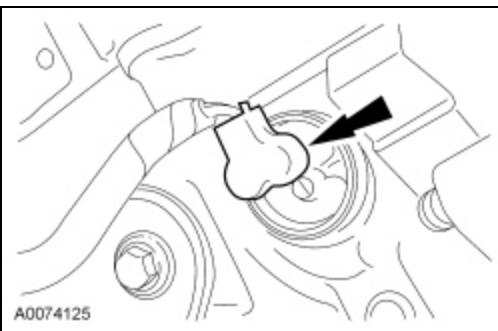


20. Install the starter. For additional information, refer to Section 303-06.

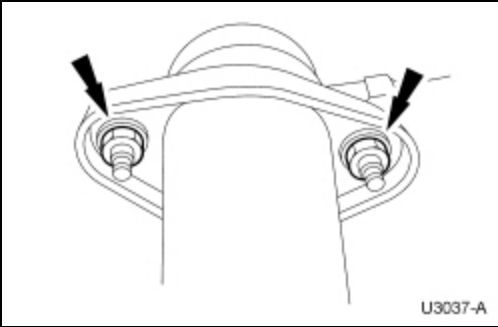
21. Attach the starter motor wiring harness pin-type retainer to the oil pan.



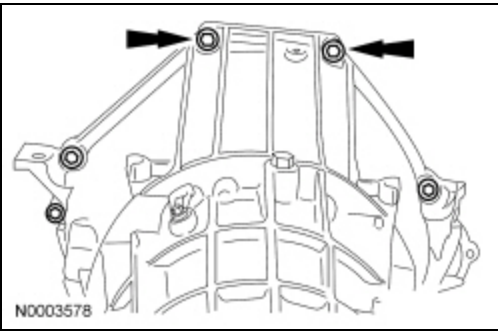
22. If equipped, connect the block heater electrical connector.



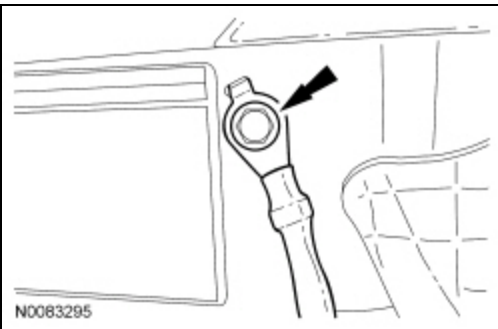
23. Install the 4 exhaust Y-pipe flange nuts (2 RH and 2 LH).
- Tighten to 40 Nm (30 lb-ft).



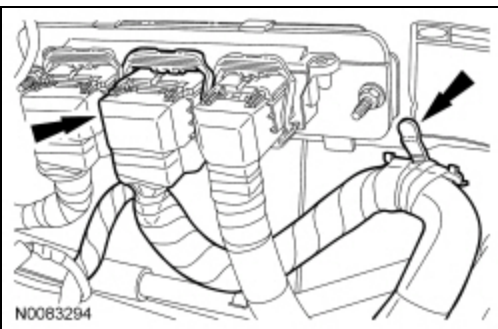
24. Position the fuel and Evaporative Emission (EVAP) tube support bracket and install the upper 2 transmission-to-engine bolts.
- Tighten to 48 Nm (35 lb-ft).



25. Position the ground strap and install the bolt.
- Tighten to 10 Nm (89 lb-in).



26. Connect the PCM electrical connector and the engine wiring harness retainer.



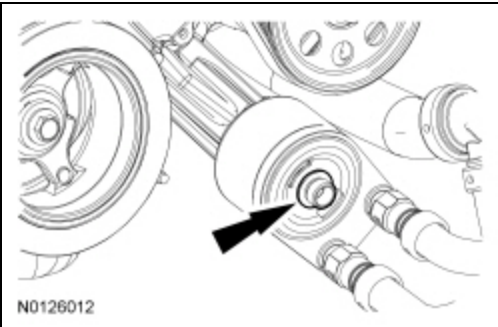
27. Connect the engine harness electrical connector and wiring harness retainer.



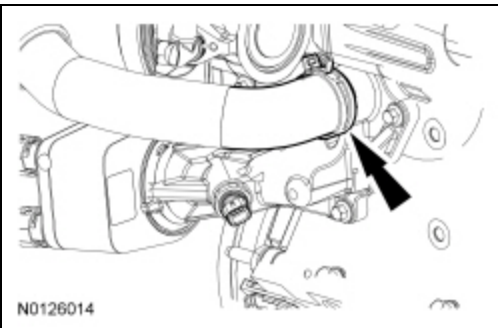
28. Connect the 2 heater hoses.

29. Position the oil cooler and install the threaded shaft.

- Tighten to 58 Nm (43 lb-ft).

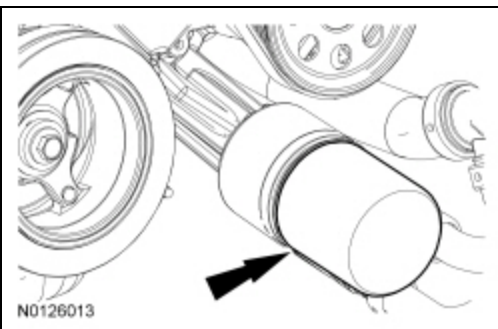


30. Connect the lower radiator hose to the oil filter adapter.



31. Install a new oil filter in 2 stages.

- Stage 1: Tighten to 5 Nm (44 lb-in).
- Stage 2: Tighten an additional 180 degrees.

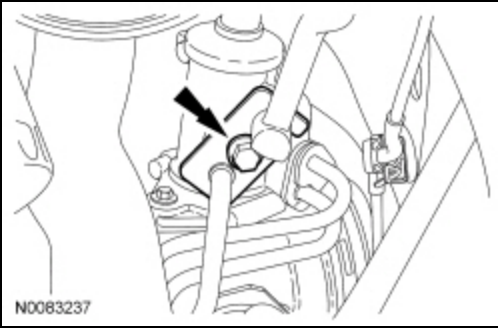


32. Install the power steering pump. For additional information, refer to Section 211-02

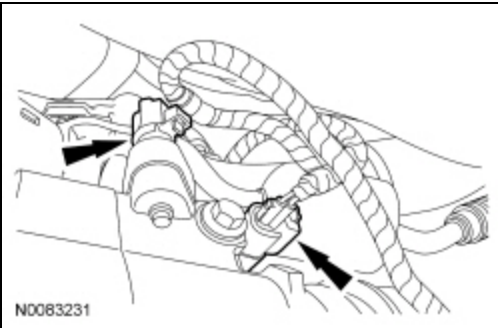
33. **NOTICE: While servicing the power steering system, care should be taken to prevent the entry of foreign material or failure of the power steering components may result.**

Install the power steering fluid return hose and install the bolt.

- Install a new o-ring seal.
- Tighten to 23 Nm (17 lb-ft).

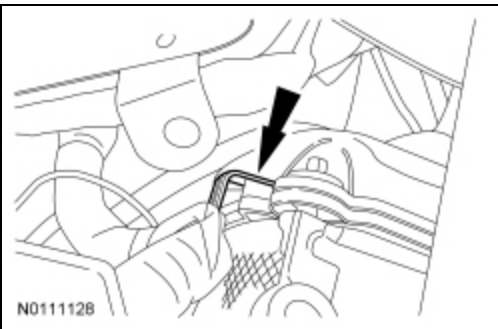


34. Connect the A/C compressor electrical connector and the wiring harness retainer.



35. Position the accessory drive belt onto the accessory drive pulleys.

36. Attach the wiring harness pin-type retainer to the RH valve cover.



37. Position the Power Distribution Box (PDB) and wiring harness onto the engine.

38. Install the cooling module. For additional information, refer to Section 303-03.

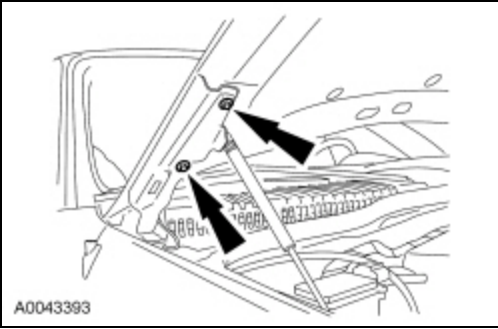
39. **NOTICE:** If the engine is repaired or replaced because of upper engine failure, typically including valve or piston damage, check the intake manifold for metal debris. If metal debris is found, install a new intake manifold. Failure to follow these instructions can result in engine damage.

Install the intake manifold. For additional information, refer to Intake Manifold in this section.

40. **NOTE:** Align the index marks made during hood removal.

Position the hood and install the 4 bolts.

- Tighten to 12 Nm (106 lb-in).



41. Fill the crankcase with clean engine oil.
42. Evacuate and charge the A/C system. For additional information, refer to Section 412-00.
43. Fill and bleed the power steering system. For additional information, refer to Section 211-00B.
44. If the engine was disassembled, use the scan tool to perform the Misfire Monitor Neutral Profile Correction procedure following the on-screen instructions.